

Decent Coaching Classes
VIII SB
Mind Your Math - VI — Answer Key

Q.1) (any five)

- (1) Centroid
- (2) $(x + 7)(x - 4) = x^2 + 3x - 28$
- (3) $p^2 + 2p - 63 = (p + 9)(p - 7)$
- (4) $a \times b = k \Rightarrow 6 \times 20 = 120 \therefore k = 120$
- (5) Rhombus
- (6) Supplementary (i.e., sum to 180°)
- (7) $x = ky \Rightarrow 15 = k \times 5 \therefore k = 3$

Q.2) (any four)

- (1) $(2a - 5b)^2 = 4a^2 - 20ab + 25b^2$
- (2) $6x^2 + 5x - 6 = 6x^2 + 9x - 4x - 6 = 3x(2x + 3) - 2(2x + 3) = (2x + 3)(3x - 2)$
- (3) $y \times x^2 = k \Rightarrow 8 \times 9 = 72 = k$. When $x = 6 : y \times 36 = 72 \therefore y = 2$
- (4) Diagonals of a rectangle bisect each other $\therefore l(AP) = l(PC) = 15/2 = 7.5$ cm
- (5) Diagonal = side $\times \sqrt{2} = 9\sqrt{2}$ cm (≈ 12.73 cm)
- (6) Centroid divides median in ratio $2 : 1 \therefore l(AG) : l(GD) = 2 : 1$. Since $l(AG) = 10$ cm, $l(GD) = 5$ cm and $l(AD) = l(AG) + l(GD) = 15$ cm

Q.3) (any three)

- (1) $(2x - 3y + 4)^2 = (2x)^2 + (-3y)^2 + (4)^2 + 2(2x)(-3y) + 2(-3y)(4) + 2(2x)(4)$
 $= 4x^2 + 9y^2 + 16 - 12xy - 24y + 16x$
 $\therefore \text{Answer} = 4x^2 + 9y^2 - 12xy + 16x - 24y + 16$
- (2) $27a^3 + 8b^3 = (3a)^3 + (2b)^3 = (3a + 2b)[(3a)^2 - (3a)(2b) + (2b)^2] = (3a + 2b)(9a^2 - 6ab + 4b^2)$
- (3) $(x^2 + 7x + 12)/(x^2 - 16) \div (x + 3)/(x - 4)$
 $= [(x + 3)(x + 4)] / [(x - 4)(x + 4)] \times (x - 4)/(x + 3)$
 $= 1$
- (4) Diagonals of a rhombus bisect each other at right angles.
Half-diagonals : $OP = 12$ cm, $OQ = 9$ cm
side = $\sqrt{(OP^2 + OQ^2)} = \sqrt{(144 + 81)} = \sqrt{225} = 15$ cm
Perimeter = $4 \times \text{side} = 4 \times 15 = 60$ cm
- (5) Opposite angles of a parallelogram are congruent and adjacent angles are supplementary.
 $\therefore \angle Y = \angle W = 65^\circ$, $\angle X = \angle Z = 180^\circ - 65^\circ = 115^\circ$
- (6) Number of men and days are in inverse variation $\therefore \text{men} \times \text{days} = k$
 $k = 12 \times 18 = 216$. For 18 men : $18 \times \text{days} = 216 \therefore \text{days} = 12$

Q.4) (any two)

- (1) Steps of construction (quadrilateral ABCD; $AB=5$, $BC=5.5$, $CD=4$, $AD=4.8$, $AC=7$, all in cm):
 - (i) Draw seg $AC = 7$ cm.

(ii) With A as centre, draw an arc of radius 5 cm; with C as centre, draw an arc of radius 5.5 cm, intersecting the first arc at point B. Draw seg AB and seg BC — this gives $\triangle ABC$.

(iii) With A as centre, draw an arc of radius 4.8 cm; with C as centre, draw an arc of radius 4 cm (on the side of AC opposite to B), intersecting at point D. Draw seg AD and seg CD — this gives $\triangle ACD$.

(iv) Quadrilateral ABCD, formed by seg AB, seg BC, seg CD and seg DA, is the required quadrilateral.

(2) Steps of construction (quadrilateral PQRS; $PQ=4.5$, $QR=5$, $RS=4$ in cm, $m\angle Q=100^\circ$, $m\angle R=80^\circ$):

(i) Draw seg $QR = 5$ cm.

(ii) At point Q, draw a ray making an angle of 100° with seg QR (with QP on the required side); mark point P on this ray such that $l(QP) = 4.5$ cm.

(iii) At point R, draw a ray making an angle of 80° with seg RQ (on the same side as P); mark point S on this ray such that $l(RS) = 4$ cm.

(iv) Join seg PS. Quadrilateral PQRS is the required quadrilateral.

(3) $\text{Speed} \times \text{time} = k$ (constant distance)

(i) $\text{Distance} = 60 \times 5 = 300$ km

(ii) $\text{Speed} \times 4 = 300 \therefore \text{speed} = 75$ km/hr

(4) Let $(3x + z) = a$ and $2y = b$.

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$\therefore (3x + 2y + z)^2 - (3x - 2y + z)^2 = 4 \times (3x + z) \times 2y = 8y(3x + z) = 24xy + 8yz$$